JOTHIKA KUMAR

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LinkedIn | Portfolio Website

About Me

Structural & Composite Design Engineer passionate about advancing lightweight structures under strict constraints. Strong technical communication and cross-functional collaboration; hands-on expertise spanning design, FEA with Python automation, failure analysis, optimization, composite fabrication, experimental testing, and validation. Committed to delivering high-performance practical solutions with engineering judgment for aerospace applications.

Education

Delft University of Technology, Netherlands **Sept 2024 - Present (Expected Aug 2026)**
Master of Science in Aerospace Engineering (with Honours) GPA: 8.7/10

- **Track:** Aerospace Structures and Materials (Design and Safety of Structures Profile)
- **Focus Areas:** Structural Analysis (FEA), Composite Structures, Crashworthy Design, Sustainability

National Institute of Technology, Tiruchirappalli **Aug 2020 - May 2024**
Bachelor of Technology in Mechanical Engineering CGPA: 9.56/10

- **Department Rank:** 2/133
- **Related Coursework:** Structural Mechanics, Strength of Materials, Mechanical Design, Computational Modeling

Work Experience

Master's Thesis, TU Delft and Collins Aerospace **Dec 2025 - Present**
Project: Failure Investigation of Curved Thermoplastic Composite Structures under Combined Loading
Supervisors: Dr. Christos Kassapoglou, Dr.-Ing. Saullo G. P. Castro, Dr. Ismet Baran

- Extracting coupons from **curved omega-section thermoplastic composite** and characterizing material behavior through off-axis tensile, longitudinal tensile, and curved-beam shear testing per **ASTM standards** to capture plastic hardening and material parameters for FEA model calibration.
- Conducting element-level testing under combined tension-bending to study unfolding failure and establish an **experimental-FEA workflow** for crashworthy composite design in aerospace and automotive applications.

Teaching Assistant, TU Delft **April 2025 - Present**

- Teaching Assistant for Computational Modeling (AE2220-II), Structural Analysis and Design (AE2135-I), Simulation, Verification and Validation (AE3212-II), and Trinity Exercise (AE4ASM105) courses offered in Aerospace Engineering.
- Contributed **300+ hours** to reviewing lecture material/assignments, grading, and mentoring **350+ students**. Demonstrated strong understanding of fundamentals and ability to communicate complex concepts clearly to students.

Internship, Collins Aerospace, Netherlands **Aug 2025 - Nov 2025**
Project: 16g Dynamic structural analysis of cabin seat system

- Developed reduced-order FE model of economy class cabin seat in Altair HyperMesh/Radioss to simulate **16g dynamic crash** loading per FAA certification standards. Correlated peak stress predictions to baseline aluminum design with **4%** error margin, enabling confidence in **Certification by Analysis**.
- **Automated** entire workflow using Python API, developing modular design framework. Reduced simulation runtime from 14 hours to **12 minutes** (**70× speedup**) using 90% fewer computational cores, enabling rapid design exploration.
- Proactively collaborated with 7 structural engineers across the global Collins network to refine model specifications and integrate design feedback into the simulation tool.

Research Project, TU Delft **Jan 2025 - June 2025**
Project: Parametric Finite Element Analysis of Structures with Inclusions *Under Prof. Christos Kassapoglou*

- Created a **parametric finite element model in Abaqus** using **Python scripting** to examine and quantify stress distribution in a composite matrix with circular fibers, and validated against analytical results with **<5% error**.
- Framework enables rapid **fiber placement assessment** without manual model setup, supporting design iterations and optimization within minutes.

Events Commissioner, Enlightness, TU Delft **Oct 2024 - Sept 2025**

- Organized **technical workshops**, **mentorship programs**, and **annual career event** facilitating networking between **100+ master's students** and industry professionals, supporting career development in aerospace engineering.

Vice President & Structures Lead, The Third Dimension Club, NIT Trichy **Aug 2021 - May 2024**

- Led team of **45+ students** across structural design, analysis, fabrication of UAVs, and composite structures for nationwide competitions. Managed cross-functional coordination between aerodynamics, structures, and avionics teams.
- Designed payload-delivery quadcopter (SAE AeroTHON '22) with Hybrid-X composite frame. Ran **topology optimization** in ANSYS, achieving **18% weight reduction** while maintaining stiffness targets. Manufactured CFRP-balsa **sandwich structure** and delivered **14-minute flight test** with successful payload delivery.

Bachelor's Thesis, National Institute of Technology, Tiruchirappalli **Dec 2023 - May 2024**

Project: Numerical Modeling of Flexible Heat Pipe for Space Applications *Under Prof. Dr. Suresh S*

- Designed and validated **ANSYS Fluent** multiphase porous-media model for flexible heat pipe. Quantified performance variation across bending angles (0–90°) with 17% correlation to experimental measurements.

Summer Research Internship, RWTH Aachen University, Germany **May 2023 - Sept 2023**

Project: Analysis of the Thermal Protection System of Reusable Launch Vehicles *Under Prof. Dr.-Ing. Kai-Uwe Schröder*

- Conducted structural and thermal **finite element analysis** of PCM-lattice composite structure; quantified convection effects for improved thermal efficiency in aerospace applications.
- Modernized **PCMLattice OpenFOAM** solver (C++/Linux) and post-processed phase-change metrics to achieve **14% reduction in melting time** on increasing gravity, demonstrating improved thermal response for space vehicle scenarios.

Skills

FEA & Simulation (Advanced): Abaqus (Standard/Explicit, Python API), Altair HyperMesh (Radioss, OptiStruct), Hyperstudy, LS-DYNA, ANSYS Mechanical, COMSOL

Programming: Python (FEA automation, material model calibration), MATLAB, C++

CAD & Post-Processing: CATIA V5, SOLIDWORKS, Fusion 360; HyperView, Hypergraph, ParaView

Standards & Certification: FAA (FAR 25), EASA (CS-25), ASTM composite testing

Composite Fabrication & Testing: Carbon/glass fiber/sandwich structures; Vacuum bagging, hot-pressing, resin infusion; Mechanical testing (tensile, compression, shear, flexure); Digital Image Correlation (DIC), Microscopy

Design & Optimization: Topology optimization, Design for Manufacturing (DFM), curve-fitting, parametric design, structural sizing, run-time reduction, weight/cost trade-offs

Soft Skills: Technical communication, stakeholder management, cross-functional team leadership, project planning, structured documentation, informed decision-making, problem-solving

Languages: English (professional), Tamil (native), Hindi (professional), French/German (beginner)

Awards

Prestigious Scholarships

- **NL Scholarship (TU Delft)** - Awarded for academic merit to cover first-year **living expenses**.
- **DAAD WISE** - Fully funded research internship in **Germany**, ranking among the top 150 students in India.
- **KC Mahindra** - Interest-free loan scholarship for **top 90 students** nationwide based on overall excellence.
- **OPJEMS** - Scholarship for **top 100 students** in India based on academic merit and leadership vision.
- **MITACS Globalink** - Selected to pursue a fully-funded research internship in **Canada**.

Academic Excellence & Recognition

- **Institute Award** - Achieved **1st rank in Mechanical Engineering** & **2nd in the university** (2020-21).
- **RSI-C IITM Summer Program** - Selected among **top 2** from 250 students, organized with **MIT & CEE**.

Publications & Conferences

Autonomous Multi-Rotor UAVs: A Holistic Approach to Design, Optimization, and Fabrication **Jan 2024**

- Presented at the International Conference on Advances in Mechanical and Aerospace Engineering **(ICAMAE)** | *Link*

Albatross: Unmanned self-sustainable glider for forest fire detection | *Link* **Sept 2022**